



CIS241

System-Level Programming and Utilities

Regular Expressions (regex)

Erik Fredericks, frederer@gvsu.edu

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Based on material provided by Erin Carrier, Austin Ferguson, and Katherine Bowers

Basics

- **Regular expression**
 - Defines a set of one or more strings of characters
- **Simple string of characters**
 - Represents itself
- **Special/metacharacters**
 - Characters that **do not** represent themselves
- **Add in special characters**
 - Match a pattern which can represent many strings



Special characters

We'll focus on the **Extended Regular Expression (ERE)** syntax

Delimiter:

- Marks beginning/end of expression
- Often `/`
- Some utilities let you use other delimiters

Char	Use	Example
<code>\</code>	escape special character	<code>a\+b</code> matches "a+b"
<code>.</code>	wildcard - match any character	<code>.ord</code> matches "word", "cord"
<code>[]</code>	character class	<code>[bB]ob</code> matches "bob", "Bob"
<code>^</code>	beginning of line	<code>^B</code> matches "B" at start of line
<code>\$</code>	end of line	<code>!\$</code> matches "!" ending line

Char	Use	Example
<code>*</code>	match 0 or more occurrences	<code>bo*</code> matches "b", "booooo"
<code>?</code>	match 0 or 1 occurrences	<code>bo?</code> matches "b", "bo"
<code>+</code>	match 1 or more occurrences	<code>bo+</code> matches "bo", "boooo"
<code>{n}</code>	match exactly <code>n</code> occurrences	<code>bo{2}</code> matches "boo"
<code>{n,m}</code>	match between <code>n</code> and <code>m</code> occurrences	<code>bo{1,2}</code> matches "bo", "boo"
<code>()</code>	group characters	<code>(da)*</code> matches "da", "dada"
<code> </code>	match next or previous	<code>hi bye</code> matches "hi", "bye"

Some examples

`a` : matches the string `a`

`a+` : matches one or more `a`s

`a*` : matches zero or more `a`s

- What does `lo+l` match?
- Where does this differ from `lo*l`?

Parentheses can group characters (called a capture group)

- What does `(ab)+` match?
- Which ones of these wouldn't match? Why?

`ab` `abab` `" "` `ba` `abababababab` `aba` `aab`

- **What does this command do?** `cat file_*`
- **What if we want that wildcard functionality in regex?**
 - `.` (a dot/period) - matches any character
- **How do we then match the same strings as the command above?**
 - `file_.*`

Applying restrictions

- Example: what would `(b.d)+` match? `b.+d?`
- What if we want to restrict the wildcard to only match vowels?
 - We use character classes `[ ]`
 - `b[aeiou]d`
 - How do `(ab)+` and `[ab]+` differ?
 - Note we can also use `tr`-like character classes:
 - `[[:digit:]] [a-z]`
 - Two sets of `[ ]`?
 - Outer: this is a character class!
 - Inner + colons: Use a `tr`-style set
 - Can also invert: `[^[:digit:]] [^0]`

Matching the *forbidden* characters

What does `foo.txt` match?

What if we want it to only match "`foo.txt`"?

- `foo\\.txt`

What does this match?

`a dog|cat`

- Matches `a dog` or `cat`. Does not match `a cat` -- the `|` operator is greedy and needs parens:
- `a (dog|cat)`

Examples

What do these match?

- `The (dog|cat) ra+n away$`
- `^bee+s*`
- `[Ll][o!]{2}[o!]*`

Examples

Create a regex to match:

ab, aba, abb, abba, abab, abbb, abaa, and nothing else

Order of operations

ERE Precedence (from high to low)			
1	Collation-related bracket symbols	[==] [::] [..]	
2	Escaped characters	\<special character>	
3	Bracket expression	[]	
4	Grouping	()	
5	Single-character-ERE duplication	* + ? {m,n}	
6	Concatenation		
7	Anchoring	^ \$	
8	Alternation		

via <https://stackoverflow.com/questions/36870168/operator-precedence-in-regular-expressions/49445993#49445993>

Using regexs

We are using extended regular expressions (ERE)

Use them with `grep` and `-E`:

- `grep -E "ab[ab]{2}" file.txt`

`grep` -> *global regular expression print*

`grep` returns any lines with a match

- To return just the match, add `-o`

Note: other commands use `/pattern/` to denote regex!

Exercises:

Which strings match the regex:

- `[^5][[:digit:]]+`
 - 12, 3, 50, 15, b0, 10000, 1050, \$10, \$4.50, 2!

Create a regex to:

- Match ab, aba, abb, abba, abab, abbb, abaa and nothing else

What does this match?

- `[[:digit:]]{10}|\([[:digit:]]{3}\)[[:digit:]]{3}-[[:digit:]]{4}`